

Scope. Companion to `4state_del_hd_ar1.tex` / `4state_del_hd.tex`: drag law $c(\bar{h})$ analytically *unknown*; a fitted curve $a(\bar{h}; \hat{\theta})$ (level + slope a') replaces *every* true- c read. Wall *position* known ($\bar{h}_d$, Δh_d , $\bar{h}_{\text{meas}}$ available).

Dual-branch (user ruling): main = AR(1) reverting-gain ($\mathbf{F}_e(4, 4) = \lambda_c$, predict reverts to $a_{x_{\text{det}}} = a(\bar{h}_d; \hat{\theta})$); companion = RW-exact ($\mathbf{F}_e(4, 4) = 1$, known- c ceiling 9.4%).

Budget: $z |\hat{a}_x - a_x|/a_x \sim 1\%$ (known- c baseline 0.4%); xy near-wall $\leq 5\%$. Supplier: θ -fit per axis ($z = Z2$, $xy = M2$; see S6). S1–S9 complete (full T1 draft); Rulings 1–5 are the design constraints, S9 the lookup.

1. Architecture ruling (dual-branch)

Per-axis augmented state $\mathbf{x} = [\delta x_1, \delta x_2, \delta x_3, a_x]^\top$; slot 4 carries two candidate dynamics. Both replace the true reads $a_{x_{\text{det}}}$, a'_x by the fitted supplier

$$a_{x_{\text{det}}}[k] = a(\bar{h}_d[k]; \hat{\theta}), \quad \hat{a}'_x[k] = \left. \frac{\partial a}{\partial \bar{h}} \right|_{\hat{\theta}}(\bar{h}_d[k]), \quad \Delta \hat{a}_x[k] = \hat{a}'_x[k] \Delta h_d[k].$$

Two slot-4 branches

$$\text{RW:} \quad a_x[k+1] = a_x[k] + \Delta \hat{a}_x[k] + \Delta a_{\text{ram}}[k], \quad \mathbf{F}_e(4, 4) = 1$$

$$\text{AR(1):} \quad a_x[k+1] = \lambda_c a_x[k] + (1 - \lambda_c) a_{x_{\text{det}}}[k] + \Delta \hat{a}_x[k] + a'_x \varepsilon[k], \quad \mathbf{F}_e(4, 4) = \lambda_c$$

The AR(1) predict reads the fitted *level* $a_{x_{\text{det}}}$ (reversion anchor) *and* the fitted *slope* $\hat{a}'_x$ (FF increment); the RW predict reads only the *slope*.

Structural fact (G1) — the reversion consumes the curve LEVEL

Define the supplier *level* error $e_{al}[k] = a_{\text{true}}(\bar{h}_d[k]) - a(\bar{h}_d[k]; \hat{\theta})$. Under AR(1), e_{al} enters the slot-4 prediction with weight $(1 - \lambda_c)$ per step. Predict/update ($\hat{a}^-[k+1] = \lambda_c \hat{a}_x[k] + (1 - \lambda_c) a_{x_{\text{det}}}[k] + \Delta \hat{a}_x[k]$, then $\hat{a}_x[k+1] = (1 - K) \hat{a}^-[k+1] + K y_2$) with constant e_{al} and $K = \text{near-wall DARE gain of the } a_x \text{ slot}$ gives the steady-state level-bias transfer

$$T := \frac{e_{a_x}^{ss}}{e_{al}} = \frac{(1 - K)(1 - \lambda_c)}{(1 - \lambda_c) + K \lambda_c} \xrightarrow{K \sim 10^{-3}} 0.99\text{--}1.00$$

i.e. a fitted *level* bias passes $\sim 1:1$ into $\hat{a}_x$ (the estimator cannot out-vote a persistent anchor bias with $K \rightarrow 0$). This is the reason the audit’s “slope-only” scope fails: the 0.4% AR(1) baseline is bought by the *level* anchor, and e_{al} is a first-class error input.

Branch trade: AR(1) consumes LEVEL (bias $T \approx 1$ but 0.4% known- c floor); RW consumes only SLOPE (no e_{al} path) but carries the documented known- c near-wall ceiling $|\hat{a}_z - a_z| = 9.4\%$ (`4state_del_hd.tex` L274 caption). Both are derived here; measured L2/L3 rungs pick the winner.

Error dynamics (supplier errors as exogenous inputs)

With slope error $e_K[k] = a'_{\text{true}}(\bar{h}_d[k]) - a'(\bar{h}_d[k]; \hat{\theta})$ (so $\Delta a_x^{\text{true}} - \Delta \hat{a}_x = e_K \Delta h_d$), the predict-only gain-error recursion for each branch:

$$\begin{aligned} \text{AR(1):} \quad e_{a_x}[k+1] &= \lambda_c e_{a_x}[k] + (1 - \lambda_c) e_{al}[k] \\ &\quad + e_K[k] \Delta h_d[k] + a'_x \varepsilon[k] \end{aligned}$$

$$\text{RW:} \quad e_{a_x}[k+1] = e_{a_x}[k] + e_K[k] \Delta h_d[k] + \Delta a_{\text{ram}}[k]$$

$a'_x\varepsilon$ (AR(1)) and Δa_{ram} (RW) are the base white drivers (q_4 ; `4state_del_hd_ar1.tex` p.5, `4state_del_hd.tex` L253). The two *new* inputs e_{al} (AR(1) only) and e_K (both) are **deterministic-dominated** — structure misfit + frozen- θ error, slowly varying, *not* white. Their carrier assignment (which Q/R entry, bridge, or offline bias table absorbs them) is *not* Q_{44} inflation; it is fixed in §3.2 (bias/variance split) and §3.3 (Ruling 1).

2. Replacement-site inventory (the true- c reads)

Every production read of the true $c(\bar{h})$ in `model/controller/motion_control_law_eq17_4state.m`, its fitted replacement, the branch it applies to, and the supplier error it then carries. Line numbers verified against the file (helper $\rightarrow$ call-site distinguished).

Station	Source (file:line)	θ -fit replacement	Branch	Error carried
$a_{x_{det}}$ level (AR(1) rever- sion anchor)	<code>local_a_x_det</code> L667– 680 (true- c read L675); consumed L531	$a(\bar{h}_d[k]; \hat{\theta})$ — fitted level from p_d	AR1	e_{al} ($\sim 1:1$, G1)
FF predict in- crement Δa_x	<code>da_x_pred</code> L306 (= $a_{det,k} - a_{det,k-1}$); consumed L531/L533	$a(\bar{h}_d[k]; \hat{\theta}) - a(\bar{h}_d[k-1]; \hat{\theta})$	both	e_K (slope, finite- diff)
y_2 known-input corr. $\sum_i \Delta a_x$	<code>sum_da_ff</code> L307 – 314 (telescoped $a_{det,k} - a_{det,k-d}$); consumed L547	$\hat{c}[k] = a(\bar{h}_d[k]; \hat{\theta}) - a(\bar{h}_d[k-d]; \hat{\theta})$	both	e_K
Q_{44} slope / <code>var_da_ram</code>	<code>K_h</code> read <code>K_h_axis</code> L320–321 ($\bar{h}_{meas}$); used L438 (AR1)/ L444 (RW)/L430 (cap)	$K_h = a'(\bar{h}_{meas}; \hat{\theta})/a$ from fit	both	e_K (2nd-order, in Q)
R_{22} delay-leak	<code>var_da_inc</code> L448 $\rightarrow$ <code>R22_delay_i</code> L468 – 472 ($d=2$)	same fitted K_h	both	e_K (2nd-order, in R)
Q_{33} randgain	L452–453 ($d=2$) / L456 ($d=1$); <i>inherits</i> true- c via <code>var_da_inc</code> (L448 $\leftarrow K_h$)	fitted K_h (no independent read)	both	2nd-order ($\leq 10^{-5}$ of Q_{33})
$\sigma_{\delta h}^2$ (cross-axis $c_\perp$ z -shape)	<code>a_perp_meas</code> L322 ($\leftarrow$ $c_\perp$ L320); <code>sigma2_dh</code> L327 (all 3 axes)	$4k_B T a_\perp(\bar{h}_{meas}; \hat{\theta}_z)$ (z -basis)	both	e_{al} of $c_\perp$; self- axis miswire $\rightarrow$ $K_{44} + 76\%$
Init block ($a_{x,init}$ / $K_{h,init}$ / DARE seed)	L176–217 (true- c read L181); $a_{x,init}$ L182, $K_{h,init}$ L183, $\sigma_{\delta h,init}^2$ L199, DARE seed L216	θ -prior (Goldman 9/16, Brenner 9/8): $a_{x,init} = a_{nom}/c(\bar{h}_{init}; \theta_0)$, $P(4,4)_{init} = (\partial a / \partial \theta)^2 \text{Var}(\theta_0)$	both	e_{al} (transient)
Optional flags	<code>use_deblur</code> L342–349 (L343 uses $a_{det,k-d}$) / <code>use_q44_ar1</code> L434–438 / <code>use_q44_cap</code> L423– 433 / <code>use_aprime_ff</code> L439–442, L522–525	$ar1$ = main branch (fit K_h); cap = fit K_h ; deblur = DISABLED (self- ref, S8); $aprime_ff$ = DISABLED (falsified, L440)	AR1 / both	e_K / —

The original plan's “slope-only 3 sites” scope is **FALSE** (G1 / INV-1): 9 stations read the true c . This table is the corrected SSOT.

3. Error dynamics + carriers

3.1 Exact error inputs (finite-difference of level evals)

The FF increment is coded as a difference of two level evaluations (`da_x_pred`, L306), not a slope read:

$$\Delta \hat{a}_x[k] = a(\bar{h}_d[k]; \hat{\theta}) - a(\bar{h}_d[k-1]; \hat{\theta}).$$

True deterministic increment $\Delta a_x^{det}[k] = a_{true}(\bar{h}_d[k]) - a_{true}(\bar{h}_d[k-1])$; the exact slot-4 FF error input is a level-error difference

$$\Delta e_{al}[k] := e_{al}(\bar{h}_d[k]) - e_{al}(\bar{h}_d[k-1]), \quad e_{al}(\bar{h}) := a_{true}(\bar{h}) - a(\bar{h}; \hat{\theta}).$$

Taylor at leading order ($' = d/dh$, $e'_{al} = e_K$, $e''_{al} = e'_K$):

$$\Delta e_{al}[k] = e_K[k] \Delta h_d[k] + \frac{1}{2} e'_K[k] \Delta h_d^2[k] + O(\Delta h_d^3), \quad e_K[k] = a'_{true}(\bar{h}_d[k]) - a'(\bar{h}_d[k]; \hat{\theta}).$$

Exact recursions (Sec. 1 form; leading order used below):

$$\text{RW: } e_{ax}[k+1] = e_{ax}[k] + \Delta e_{al}[k] + \Delta a_{ram}[k]$$

$$\text{AR(1): } e_{ax}[k+1] = \lambda_c e_{ax}[k] + (1 - \lambda_c) e_{al}(\bar{h}_d[k]) + \Delta e_{al}[k] + a'_x \varepsilon[k]$$

AR(1) carries the level *twice*: $(1 - \lambda_c) e_{al}$ (reversion anchor, $O(e_{al})$) + Δe_{al} (FF, $O(e_K \Delta h_d) \ll e_{al}$). RW carries only Δe_{al} ; no $(1 - \lambda_c) e_{al}$ path. This is the branch split of Sec. 1 — RW's level bias enters only through the FF slope-difference, AR(1)'s through the anchor.

3.2 Deterministic/stochastic split + carriers

$$e_{al}(\bar{h}, k) = \underbrace{b_{lvl}(\bar{h})}_{\text{struct. misfit + frozen-}\theta} + \underbrace{s_{lvl}[k]}_{(\partial a / \partial \theta)^\top \delta \hat{\theta}}, \quad e_K(\bar{h}, k) = b_K(\bar{h}) + s_K[k]$$

Jacobian map (RLS posterior P_θ , $\delta \hat{\theta} = \hat{\theta} - \theta^*$, zero-mean):

$$\text{Var}(s_{lvl}) = (\partial a / \partial \theta)^\top P_\theta (\partial a / \partial \theta), \quad \text{Var}(s_K) = (\partial a' / \partial \theta)^\top P_\theta (\partial a' / \partial \theta).$$

Carrier rule (bias/variance split). *Deterministic* b_{lvl} , b_K (functions of $\bar{h}$, slowly varying, NOT white): carrier $\in \{\text{offline bias table } e_{ss}(\bar{h}) \mid \text{deterministic bridge (subtract known lag) \mid gate (trough, } y_2 \text{ off)}\}$. **Never** Q .
Stochastic s_{lvl} , s_K (RLS noise, white at loop rate): carrier = Q_{44} additive via Jacobian — RW: $Q_{44} += \text{Var}(s_K) \Delta h_d^2$ (relative form, §3.6); AR(1): $Q_{44} += (1 - \lambda_c)^2 \text{Var}(s_{lvl})$ (§3.5).

3.3 Q_{44} inflation of the e_K ramp — rejected

Slot-4 input $u[k] = e_K[k] \Delta h_d[k]$ is a deterministic ramp, correlation time $\tau_c \approx \text{quarter-period} = f_s / (4f) = 1600 / 4 = 400$ steps. Its only white (unpredictable) content is the one-step innovation

$$\Delta u[k] = u[k] - u[k-1] \approx e_K \Delta^2 h_d[k], \quad \Delta^2 h_d = \ddot{h}_d T_s^2 = -A_h \omega^2 T_s^2 \sin \omega t,$$

so the white-to-signal ratio is set purely by the sampling kinematics

$$\frac{q_{eq}}{\text{Var}(u)} \approx \left(\frac{\Delta u}{u} \right)^2 = (\omega T_s)^2 = \left(\frac{2\pi}{1600} \right)^2 = 1.54 \times 10^{-5}.$$

Against the base driver $Q_{44} = \text{Var}(\Delta a_{ram})$ (z , trough) this gives

$$q_{eq} \leq 6.4 \times 10^{-5} Q_{44} \quad (z)$$

Ruling 1: Q_{44} inflation is an **invalid carrier** for $e_K \Delta h_d$ (white content $< 10^{-4}$ of Q_{44}). The ramp is $\sim$ pure low-frequency: its effect is a *bias*, not a variance. Use the ramp-lag table (§3.4).

6.4×10^{-5} (audit C1-02 numerical eval) = $(\omega T_s)^2$ kinematic factor $\times \text{Var}(u)/Q_{44}|_{trough}$ ($O(4)$); the $(\omega T_s)^2$ scaling is the load-bearing structural result — agreement, not disagreement.

3.4 Honest carrier: ramp-lag bias

RW slot-4 error is restored only by the KF gain K_{44} ($\mathbf{F}_e(4, 4) = 1$):

$$e_{ax}[k+1] = (1 - K_{44}) e_{ax}[k] + u[k], \quad u = e_K \Delta h_d = r \hat{a}'_x \Delta h_d, \quad r := e_K / \hat{a}'_x,$$

quasi-static (u const):

$$e_{ss}^{RW}(\bar{h}) = \frac{r \hat{a}'_x(\bar{h}) \Delta h_d}{K_{44}(\bar{h})}$$

AR(1) adds the reversion pole $(1 - \lambda_c)$: $e_{ax}[k+1] = (1 - K_{44})[\lambda_c e_{ax}[k] + u[k]]$, so

$$e_{ss}^{AR(1)} = \frac{(1 - K_{44}) r \hat{a}'_x \Delta h_d}{(1 - \lambda_c) + K_{44} \lambda_c}, \quad \frac{e_{ss}^{AR(1)}}{e_{ss}^{RW}} \approx \frac{K_{44}}{1 - \lambda_c} \approx 8 \times 10^{-3}.$$

The reversion suppresses the FF ramp-lag $\sim 125\times$ vs RW (trough): the slope-error ramp is an **RW-branch** problem. (AR(1)'s price is instead the LEVEL bias $(1 - \lambda_c)e_{al}$, §3.5, transfer $T \approx 1$.)

Sinusoid rejection u reverses sign each half-cycle; the leaky integrator $H(z) = 1/(z - (1 - K_{44}))$ attenuates the quasi-static peak by

$$A_{rej}(\bar{h}) = K_{44} |H(e^{j\omega T_s})| = \frac{1}{\sqrt{1 + (\omega T_s / K_{44})^2}}$$

evaluated at the bias-peak *flank* (where $|\Delta h_d|$ is max — not the trough, where $\Delta h_d \rightarrow 0$). Flank $K_{44} \approx 5 \times 10^{-3}$: $A_{rej} \approx 0.80$ (reproduces audit T1 step 4). The trough $K_{44} = 2.4 \times 10^{-3}$ would give 0.52, but $\Delta h_d \approx 0$ there so it is not the operative point. (See final message for the flag: the audit's $\times 0.80$ holds only at the flank K_{44} ; a naive trough eval gives 0.52.)

Schmidt consider alternative (optional). Carry θ as a *consider* parameter — propagate fixed $P_{\theta\theta}$ and cross $P_{x\theta}$ in the covariance, no θ state, no gain on θ ; inflates slot-4 P by $(\partial a / \partial \theta) P_{\theta\theta} (\partial a / \partial \theta)^\top$. Captures s_{lvl}, s_K (stochastic) but not b_{lvl}, b_K (still bias table). Cost: a $4 \times n_\theta$ cross block per axis per step, no control-path change.

3.5 AR(1) level term in Q_{44}

Only the *stochastic* s_{lvl} is Q -able; the deterministic b_{lvl} is a bias (transfer $T \approx 1$, G1) that goes to the table:

$$Q_{44} += (1 - \lambda_c)^2 \text{Var}(s_{lvl}) = (1 - \lambda_c)^2 (\partial a / \partial \theta)^\top P_\theta (\partial a / \partial \theta)$$

The audit's " $Q_{44} += (1 - \lambda_c)^2 \text{Var}(e_{al})$ " is read as the *stochastic* component only; b_{lvl}^2 is not a variance and must not enter Q — it surfaces as the 1:1 level bias, the un- Q -able cost of the AR(1) anchor.

3.6 Relative-form rule + far-field checklist

Ruling 2 (relative-form rule, commit 121f86a). Every new Q/R term scaling with Δh_d^2 or $\dot{h}$ must be parameterized **relative** to $\hat{a}'_x(\bar{h})$ (i.e. $\propto \hat{a}'_x(\bar{h}) \times \text{dimensionless}$), so it auto-deflates far field ($\sim 8.6 \times 10^{-4}$). A frozen absolute coefficient (e.g. $\text{Var}(e_K)$ pinned near-wall) over-inflates far-field Q_{44} by 38–65 $\times$ (@ $\bar{h}=22$) and **diverges in closed loop while looking fine open-loop**.

Checklist (each new $\Delta h_d^2/\dot{h}$ term): (i) $\bar{h} \rightarrow \infty$ limit of the term $\rightarrow 0$ (since $\hat{a}'_x \rightarrow 0$, $K_h \rightarrow 0$); (ii) term/ Q_{44}^{base} bounded over the whole trajectory (no $> 10 \times$ excursion); (iii) named test: closed-loop far-field descent segment, `fig_L2_C1_q44_farfield` — any such term $> 10 \times$ its near-wall design value $\Rightarrow$ STOP.

3.7 Numeric anchors

K_{44} (z DARE): 2.4×10^{-3} (trough) / 5.8×10^{-9} (@ $\bar{h}=22$). $r_z \geq 3\% \Rightarrow$ breaks z 1% budget.

Term ($ \hat{a}_x - a_x /a_x$)	$\bar{h}=1.2$	$\bar{h}=1.5$	$\bar{h} \geq 3.4$	Carrier
b_{lvl} level bias (z, Z2)	0.94%	$\sim 0.7\%$	$\rightarrow 0$	bias table; $\text{AR}(1) \times T \approx 1$
b_{lvl} level bias (xy, M2)	1.16%	$\sim 0.9\%$	$\rightarrow 0$	bias table
e_{ss}^{RW} ramp (z, $r=1.44\%$)	0.23–0.33%	$< 0.2\%$	$\rightarrow 0$	offline $e_{ss}(\bar{h}) \times A_{rej}$
e_{ss}^{RW} ramp (xy, $r \approx 16\%$)	3–4%	2–3%	$\rightarrow 0$	offline; budget $\leq 5\%$
$Q_{44} += (1 - \lambda_c)^2 \text{Var}(s_{lvl})$	rel.	rel.	$\rightarrow 0$	relative form (§3.6)

Level bias (b_{lvl}) and slope-ramp (e_{ss}) both auto-deflate far field ($\hat{a}'_x \rightarrow 0$); the far-field $K_{44} = 5.8 \times 10^{-9}$ does *not* blow up $e_{ss} = r \hat{a}'_x \Delta h_d / K_{44}$ because the numerator $\hat{a}'_x \Delta h_d \rightarrow 0$ faster — the relative-form rule (§3.6) is what guarantees this.

4. y_2 known-input correction

4.1 Model-exact telescoped level correction

Base y_2 correction is the known telescoped increment sum (`sum_da_ff`, L307–314, consumed L547):

$c[k] = \sum_{i=1}^d \Delta a_x[k-i] = a_{x_{det}}[k] - a_{x_{det}}[k-d]$. Under θ -fit it becomes the fitted level difference

$$\hat{c}[k] = a(\bar{h}_d[k]; \hat{\theta}) - a(\bar{h}_d[k-d]; \hat{\theta})$$

Like the FF (§3.1) this is a difference of two fitted *level* evals — no a' read, exact within the fitted model (no Taylor step). Corrected $y_2 = a_{xm} + \hat{c}$ measures $a_x[k]$ with residual

$$r_2[k] = \underbrace{n_a[k] - \sum_{i=1}^d \Delta a_{ram}[k-i]}_{\text{base } r_2} - e_c[k],$$

$$e_c[k] = c_{true}[k] - \hat{c}[k] = \Delta_d e_{al}[k] := e_{al}(\bar{h}_d[k]) - e_{al}(\bar{h}_d[k-d]) \approx \sum_{i=1}^d e_K[k-i] \Delta h_d[k-i].$$

Same level-error-difference structure as the FF, now over a d -step window.

4.2 No new R_{22} term (and no double-count)

e_c is deterministic-dominated (the b_K ramp of §3.3, seen through a d -step window); its white content is negligible. A conservative all-as-variance estimate at the flank is $\lesssim 10^{-7} R_{22}$, within the audit bound

$$\text{Var}_{\text{white}}(e_c) \leq 1.5 \times 10^{-6} R_{22} \quad (\text{near wall, C1-06}) \Rightarrow \text{no new } R_{22} \text{ term}$$

No double-count. The deterministic part of e_c is the *same* b_K/b_{lvl} curve misfit already carried by the §3.4 slot-4 bias table — one physical error, seen once in the predict (FF, §3.1) and once in y_2 (e_c). Do **not** add both a bias-table entry *and* an R_{22} term for it.

4.3 Q – R cross (documentation-debt payoff)

r_2 shares noise with the slot-4 process input, so $E[q_4 r_2] \neq 0$ in general. Two crosses:

$$\text{pre-existing: } E[q_4[k] r_2[k]] = \begin{cases} 0, & \text{AR(1): } q_4 = a'_x \varepsilon[k] \text{ white, } r_2 \text{ past-only} \\ -\sum_{i=1}^d R_{\Delta a_{ram}}(i), & \text{RW: } |\cdot| \lesssim (1 - \lambda_c) Q_{44}^{RW} \end{cases}$$

θ -fit (shared s_K): $E[\Delta e_{al}[k] (-e_c[k])] = 0$ (s_K white, disjoint lags), $\leq \rho_{s_K} \sqrt{Q_{44}^{s_K} R_{22}^{e_c}}$ otherwise.

Normalized by $\sqrt{Q_{44} R_{22}}$ all are $\ll 1$ ($Q_{44} \ll R_{22}$ near wall). **Standing assumption** (missing from `4state_del_hd.tex`): diagonal $\mathbf{R}$, $E[\mathbf{q} \mathbf{r}^\top] \approx 0$, with the bounds above. AR(1) is *exactly* cross-free — one more reason it is the cleaner base.

4.4 De-blur is the alternative implementation

With `use_deblur` the code skips $\hat{c}$ (L544–548) and instead normalizes a_{xm} by $a(\bar{h}_d[k-d]; \hat{\theta})$ (L342–349): $\hat{\kappa} = \text{EWMA}[(\delta x_r^2 - C_n \sigma_n^2)/(C_{dpmr} 4k_B T a_{x_{det}}[k-d])]$, $a_{xm}^{db} = \hat{\kappa} a_{x_{det}}[k]$ — the *same* d -step alignment, performed inside the measurement rather than as an additive correction. Scale cancels in the $\hat{\kappa}$ ratio (weak self-reference); **DISABLED pending the §8 independence analysis** (per Sec. 2 table).

Numeric anchor. z , flank ($r=1.44\%$ Z2 slope, $d=2$): $|e_c| \sim 10^{-4} a_x \ll \sqrt{R_{22}} \approx 0.45 a_x$ (the 44–47% per-step a_{xm} chi-sq floor). e_c sits ~ 3 orders below the measurement noise — invisible to y_2 .

5. Gate-off propagation

5.1 Setup

Guard G3 ($\bar{h}_{\text{meas}} < h_{\text{safe}} = 1.5$, code L477) sets $R_{22} \rightarrow R_{OFF} = 10^{10}$ (L482): the EKF runs y_1 -only, slot 4 predict-only ($K_{44} \rightarrow 0$). On the $\text{osc}_{1\text{Hz}}$ trough passage this is $N \approx 383$ consecutive steps/pass with y_2 off. Moreover at the trough $F_{dx} \rightarrow 0$ (velocity reversal $\Rightarrow f_d \rightarrow 0$), so $\mathbf{F}_e(3, 4) = -F_{dx} \rightarrow 0$ and y_1 carries *no* info on slot 4 (C1-03): during trough gate-off slot 4 is **fully unobservable**. Only the model bounds it.

5.2 Error propagation (both branches)

Under the Phase-1 G0 ruling the supplier is *also* gated, so on $\bar{h} \in [1.2, 1.5)$ the fitted curve is pure extrapolation: level $b_{vl}^{\text{ex}}(\bar{h})$, slope $e_K^{\text{ex}} = r^{\text{ex}} \hat{a}'_x$ (both = L0 gated-fit columns).

$\text{RW: } e_{a_x}(t) = e_{a_x}(k_0) + \int_{h_d(k_0)}^{h_d(t)} e_K dh' + \sum_{k_0}^t \Delta a_{ram}$ $\text{AR(1): } e_{a_x}(t) \rightarrow b_{vl}^{\text{ex}}(\bar{h}_d(t)) \text{ within } \frac{1}{1-\lambda_c} \approx 3 \text{ steps}$

RW **integrates** e_K (sign-coherent on descent; no restoring $\Rightarrow$ it also random-walk-drifts). AR(1) **locks onto the extrapolation level** b_{vl}^{ex} (reversion needs no y_2): model-based, so it bounds slot 4 even when fully unobservable — the one mechanism alive in the trough.

5.3 Peak error at the trough

AR(1): $|e_{a_x}^{\text{peak}}| = |b_{vl}^{\text{ex}}(1.2)| = 1.83\%$ (z , Z2 gated), 5.40% (xy , M2 gated).

RW: the descent integral (audit C1-03 form)

$$e_{a_x}^{\text{peak, RW}} = \int_{1.5}^{1.2} e_K dh' \approx r(a(1.5) - a(1.2)), \quad \frac{a(1.5) - a(1.2)}{a(1.2)} = \frac{c_{\perp}(1.2)}{c_{\perp}(1.5)} - 1 = \frac{6.35}{4.34} - 1 = 0.463$$

so z : $e_{a_x}^{\text{peak, RW}}/a \approx 0.46r$ (of local trough a). For the gated z slope error $r \sim 1\text{--}2\%$: 0.5–0.9%. xy : integrated $\sim 15\%$ (audit; M2 gated slope 21–67%). (**Flag:** the audit’s “ $\sim 1\%$ of local a per 1% slope” implies sensitivity ~ 1.0 ; my closed form gives 0.46 from $c_{\perp}(1.2)/c_{\perp}(1.5)$ — a $\sim 2\times$ gap, reconcilable if r^{ex} grows toward the trough so the integral weights the deep region. See final message; resolve at L2.)

The round trip cancels the deterministic integral by re-entry ($\int_{1.5}^{1.2} + \int_{1.2}^{1.5} \approx 0$); RW’s error *at re-entry* is $e_{a_x}(k_0) +$ random-walk drift $\sqrt{N} \sigma_{\Delta a_{ram}}$, not the integral.

5.4 $P(4, 4)$ deterministic floor

With y_2 off, P_{44} grows only by Q_{44}/step (RW: $P_{44+} = Q_{44}$; AR(1): $P_{44} \rightarrow q_4/(1-\lambda_c^2)$, bounded) — but the deterministic extrapolation error is *not* in Q_{44} (§3 carrier rule), so P_{44} understates the true slot-4 uncertainty in the trough.

Ruling 3 (gate-off P floor). While G3 active, add a <i>deterministic</i> $P(4, 4)$ floor (a floor, not Q): $P_{44}^{\text{fl}}(k) = [b_{vl}^{\text{ex}}(\bar{h}_d[k])]^2 \quad (\text{AR(1)})$ $= [\int_{h(k_0)}^{h(k)} e_K dh']^2 \quad (\text{RW})$ Remove it on gate re-entry. It represents the bias-envelope the filter is blind to, preventing overconfident fusion at the first post-gate y_2 sample.
--

5.5 Gate re-entry transient

At $\bar{h} \uparrow 1.5$, y_2 returns; the first correction is $\Delta \hat{a}_x = K_{44}^{re}(e_{a_x} + r_2)$, $K_{44}^{re} = P_{44}^{\text{fl}}/(P_{44}^{\text{fl}} + R_{22})$. With $\sqrt{R_{22}}/a \approx 0.45$ (the 44–47% a_{xm} chi-sq floor, $R_{22}/a^2 \approx 0.2$) the recovery time is

$$\tau_{re} \approx \frac{1}{K_{44}^{re}} \approx \frac{R_{22}/a^2}{(\text{peak}/a)^2} = \frac{0.2}{(\text{peak}/a)^2} \quad (P_{44}^{\text{fl}} \ll R_{22})$$

So the floor speeds recovery from $\gg 10^4$ steps (no floor: P_{44} tiny $\Rightarrow$ overconfident, clings to bias) to R_{22} -limited τ_{re} . **Flag:** τ_{re} is $O(1\text{--}10)$ steps *only* for large bias (xy RW 15% $\Rightarrow \sim 10$); for small bias (z 1–2% $\Rightarrow \sim 500\text{--}2000$) recovery stays a_{xm} -SNR-limited, *not* “ $O(1)$ step”. Crucially, **AR(1) needs no re-entry correction:** its reversion self-corrects $e_{a_x} \rightarrow b_{lvl}(\bar{h})$ (in-band) in ~ 3 steps as $\bar{h}$ climbs, independent of y_2 — the reversion *is* the gate-off floor. Re-entry recovery is an RW-branch problem.

5.6 Supplier side (frozen extrapolation)

Under Phase-1 G0 the RLS sees no data on $[1.2, 1.5]$; the extrapolation curves b^{ex} are **frozen** properties of (basis + training band) = exactly L0’s gated-fit columns. *Forward pointer:* if T3-7’s Phase-2 flip (β -corrected near-wall training) lands, the penalty collapses to the dwell-trained numbers (z 1.83 $\rightarrow$ 0.94%, xy 5.40 $\rightarrow$ 1.16%); the S5 formulas stay valid with the new (smaller) b curves.

5.7 Numeric anchors

$\lambda_c=0.7$, $N=383$, $R_{22}/a^2 \approx 0.2$, L0 gated-fit levels.

Quantity	z AR(1)	z RW	xy AR(1)	xy RW
peak trough $ e_{a_x} /a_x$	1.83%	0.5–0.9%	5.40%	$\sim 15\%$
$P_{44}^{\text{fl}}/a_x^2 = (\text{peak})^2$	3.4×10^{-4}	$\sim 5 \times 10^{-5}$	2.9×10^{-3}	2.3×10^{-2}
recovery mechanism	reversion	$y_2 + \text{floor}$	reversion	$y_2 + \text{floor}$
recovery time (steps)	~ 3	~ 2000	~ 3	~ 10
recovery, no floor (RW)	—	$\gg 10^4$	—	$\gg 10^4$

AR(1) recovers by reversion ($\sim 1/(1-\lambda_c)$ steps) regardless of y_2 or peak size — the reversion is the natural gate-off floor. RW recovery is R_{22} -limited ($\tau_{re} \approx 0.2/\text{peak}^2$): fast only when the bias is large (xy). This is the decisive gate-off argument for AR(1).

6. θ -RLS supplier spec

6.1 Training-stream data model

The supplier trains on a_{xm} , which carries three corruptions on top of a_{true} :

$$a_{xm}[k] = \underbrace{m(g[k])}_{\text{g-loop}} \cdot \underbrace{(1 + \beta(\bar{h}))}_{\text{h-blur}} \cdot a_{true}(\bar{h}[k-d-L_{eff}]) + \underbrace{\nu[k]}_{\text{chi-sq floor}}$$

- $m(g) = C_{dpmr}(\lambda_{eff})/C_{dpmr}(\lambda_c)$, $\lambda_{eff} = 1 - g(1 - \lambda_c)$, $g := a_{true}/\hat{a}_x$ — the g-loop scale (C1-08 + cross-session 2026-07-07). $m(1)=1$; $dm/dg|_1 = -0.41$.
- $\beta(\bar{h})$ — EWMA h-blur bias ($\sim 63\%$ of the measured +11% at trough; G3, T3-1).
- $L_{eff} = (1 - a_{cov})/a_{cov} = 19$ steps — EWMA window-centre lag (C1-11).
- $\nu[k]$ — chi-sq floor, rel. std 44–47%/step, $\tau_{int} \approx 39$ steps (C1-10).

6.2 RLS specification (per axis)

Replaces the exploratory hand-set scalar KF of C1-10 ($R=(0.42a)^2$, $\tau=20$, which under-counts $\text{Var}(e_\theta)$ by $\sim 2.2\times$ since the true $\tau_{int}=39$).

Regressor pairing: $(\bar{h}_d[k-d-L_{eff}], a_{xm}[k])$ — the window-centre correction of §3/C1-11 (pair the measurement to the height at its EWMA window centre, not $\bar{h}[k]$).

Basis + priors (G0), as Bayesian pseudo-observations:

$$z \text{ (Z2)} : c = 1 + \frac{\theta_1}{h-1} + \frac{\theta_2}{h}, \quad \theta_1 + \theta_2 \sim N\left(\frac{9}{8}, \sigma_{pr}^2\right) \text{ (Brenner; pins collinear, corr 0.94)}$$

$$xy \text{ (M2)} : a = a_{nom}(1 - b_1u - b_3u^3), \quad u := \frac{1}{h}, \quad b_1 \sim N\left(\frac{9}{16}, \text{soft}\right) \text{ (Faxén)}, \quad b_3 \rightarrow 0 \text{ (corr 0.97)}$$

Fallbacks Z1 rel-LS / M1 stay defined as ladder rungs.

Measurement variance (the derived chi-sq floor, bound to `build_eq17_6state_constants`):

$$R_{RLS}[k] = K_{var} IF_{eff} (\hat{a}_x + \xi)^2$$

In-band training discipline: train only when G3 off and $\bar{h}_d \leq h_{train}$ (~ 4 –5). The info weight $(\partial a / \partial \theta)^2 / R_{RLS}$ decays far-field naturally (numerator $\rightarrow 0$ as $\bar{h} \rightarrow \infty$); the hard cap h_{train} additionally blocks far-band data, which would otherwise flip the sign of the 2nd parameter (L0 band-shift: $\theta^*(\text{band})$ drift).

Forgetting $\leftrightarrow \text{Var}(e_\theta)_{ss}$ (scalar, persistent excitation $\phi = \partial a / \partial \theta|_{\text{train}}$):

$$P_{\theta,ss} \approx \frac{(1 - \lambda_f) R_{RLS}}{\phi^2}, \quad \text{Var}(s_{lvl}) = \left(\frac{\partial a}{\partial \theta}\right)^2 P_{\theta,ss}, \quad \text{Var}(s_K) = \left(\frac{\partial a'}{\partial \theta}\right)^2 P_{\theta,ss}$$

(the Jacobian map of §3.2 into Q_{44}). Smaller $1 - \lambda_f \Rightarrow$ smaller supplier noise, but slower tracking of $\theta^*(\bar{h})$ — and, crucially, a smaller anchor-loop gain (§6.3), which is the binding constraint.

6.3 Anchor-loop design constraint

The RLS anchor $a_{\hat{x}_{det}} = a(\bar{h}_d; \hat{\theta})$ closes the g-loop through *physics*: $\hat{a}_x \rightarrow f_d \rightarrow \delta x \rightarrow \hat{\sigma}_{\delta x_r}^2 \rightarrow a_{xm} \rightarrow \hat{\theta} \rightarrow a_{\hat{x}_{det}} \rightarrow \text{predict}$ (cross-session 2026-07-07, 5-layer verified). The anchor moves per step by $(\partial a / \partial \theta) K_\theta[k]$ per unit innovation; with forgetting the steady RLS gain gives, at matched training/operating height,

$$\alpha_{eq} = \frac{\partial a / \partial \theta|_{\bar{h}_d[k]}}{\partial a / \partial \theta|_{\text{train}}} (1 - \lambda_f) |P_{tilt}(\bar{h})| \approx (1 - \lambda_f) |P_{tilt}(\bar{h})|.$$

Two structural suppressors vs the raw a_{m_det} anchor

- **(i) Scale-absorption blocking:** a global scale m on the data cannot be represented by the basis (a_{nom} fixed); by the G6 mechanism it maps to a bounded *tilt* ($d\theta=+2.56\epsilon$ dwell-weighted), not 1:1 scaling. The residual projection at the operating height is $|P_{tilt}| \approx 0.9$ (-0.92 trough, sign-flipped $\Rightarrow$ stabilizing; $+0.88$ far). *Weak* suppressor ($\sim 10\%$).
- **(ii) Memory:** the RLS integrates the whole history $\Rightarrow$ per-step gain $\rightarrow (1 - \lambda_f)$, vs the a_{m_det} EWMA $\alpha=0.005$ ($P_{tilt}=1$, only $1.6\times$ below the cliff). This is the *dominant* suppressor — *but only* if λ_f meets the constraint; a fast-forgetting RLS recreates the cliff.

Ruling 4 (anchor-loop safety, not acyclicity). Algorithmic acyclicity is **not** the safety argument (cross-session falsification: the loop closes through physics; noise-triggered escape at cliff $\alpha^*=0.008$, seed-dependent). The argument is loop-gain below the cliff with margin $M \geq 4$ (justified by the 46% chi-sq fluctuation + seed-dependence: $0.007 \rightarrow 0/5$, $0.008 \rightarrow 1/5$, $0.020 \rightarrow 5/5$):

$$\alpha_{eq} = (1 - \lambda_f) |P_{tilt}| \leq \frac{\alpha^*}{M} = \frac{0.008}{4} = 0.002 \quad \Rightarrow \quad \boxed{\lambda_f \geq 0.999}.$$

Full physical-loop treatment (mean part = C1-08 fixed point $\lambda_{eff}=1-g(1-\lambda_c)$, $dm/dg=-0.41$): §8.

6.4 Warm-up handoff

Before θ convergence the FF runs the A-floor $\hat{a}'_x=0$ (§7): the slot-4 FF is off, so the g-loop is not yet closed. Enable the fitted slope only when the RLS slope uncertainty falls below the target (consistent with §7):

$$(\partial a' / \partial \theta)^2 P_\theta < (r_{target} \hat{a}'_x)^2.$$

The soft prior collapses P_θ fast, bounding the early K_θ -transient in α_{eq} .

6.5 Forbidden inputs

The RLS must never read: $\hat{a}_x$; a_{ctrl} ; EKF innovations (e_{y_1}, e_{y_2}); fit-normalized de-blur output ($\hat{\kappa} a_{x_{det}}$, self-referential). **Plus (physical-loop):** any quantity whose statistics depend on tracking quality — $\hat{\sigma}_{\delta x_r}^2$ may enter *only* through a_{xm} (no second entry point), else the loop gains a bypass around P_{tilt} . *Future (out of MVP):* p_{md} ($\delta \bar{x}_m$, the LP mean) carries a $(1-g) D[k]$ first-moment term — a *signed* g-observation channel (parallel-session discovery), a candidate de-confounder. Forward-ref S8/T3.

6.6 Numeric anchors

Priors: $z \theta_1 + \theta_2 \sim N(1.125, \sigma_{pr}^2)$; $xy \ b_1 \sim N(0.5625, \text{soft})$, $b_3 \rightarrow 0$. $R_{RLS} \propto (\hat{a}_x + \xi)^2$: smaller (abs.) at trough ($\hat{a}$ small), larger far. $|P_{tilt}| \approx 0.9$, $\alpha^*=0.008$, $M=4$.

λ_f	$1 - \lambda_f$	α_{eq} $(1 - \lambda_f) P_{tilt} $	= margin α_{eq} / α^*	verdict	$\text{Var}(s_{lvl}) @ \text{trough}$
1	0	0	0	PASS [†]	$\rightarrow 0$ (frozen)
0.9999	10^{-4}	9×10^{-5}	0.011	PASS	$\sim 10^{-4} (\hat{a}\xi\text{-set})$
0.999	10^{-3}	9×10^{-4}	0.11	PASS	$\sim (1 - \lambda_f) R_{RLS}$
0.99	10^{-2}	9×10^{-3}	1.13	FAIL	(above cliff)

Constraint $\alpha_{eq} \leq \alpha^* / M = 0.002$ (margin ≤ 0.25) $\Rightarrow \lambda_f \geq 0.999$. $\lambda_f=0.99$ sits *above* the cliff (margin 1.13) — diverges. [†] $\lambda_f=1$ is safest for the loop but freezes $\theta^*(\bar{h}) \Rightarrow$ band-dependent bias (C1-10); the working range

is $\lambda_f \in [0.999, 1)$. Note $\text{Var}(s_{lvl}) \propto (1 - \lambda_f)$ trades *against* loop safety and tracking — all three want the same $\lambda_f \gtrsim 0.999$.

7. Init + warm-up

Replaces every true- c init read (CR-3: $\mathbf{a_x_init}$ L182, $\mathbf{K_h_init}$ L183, $\sigma_{\delta h, init}^2$ L199, DARE P seed L216) with a θ -prior evaluation. The trajectory gives a clean far-field warm-up: hold 0.5 s at $\bar{h} \approx 22 \rightarrow$ descend 1 s $\rightarrow$ osc.

7.1 θ prior init values

Centers from S6 universal physics: z (Z2) $\theta_1 + \theta_2 = \frac{9}{8}$ (Brenner), xy (M2) $b_1 = \frac{9}{16}$ (Faxén), $b_3 = 0$. **Split decision** (z):

$$\boxed{\theta_{1,0} = \frac{9}{8}, \quad \theta_{2,0} = 0}$$

Far-field data constrains only the *sum* ($\frac{1}{\bar{h}-1}$ and $\frac{1}{\bar{h}}$ are collinear as $\bar{h} \rightarrow \infty$, corr 0.94); the split is resolved only near the wall, unknown pre-data, so $\theta_{2,0}=0$ is the neutral choice. The L0-informed split ($\theta_1^*=1.062, \theta_2^*=0.116$) is **rejected**: it would be read from the true c (scope violation) — literature gives the *sum* $\frac{9}{8}$ (universal), not the split.

Prior widths (cover the true value, bound the early K_θ -transient, §6.4): the true θ^* sits ~ 0.05 – 0.12 from centers (z $\theta_1^* + \theta_2^* = 1.178$ vs 1.125 , $+4.7\%$; xy $b_1^* = 0.539$ vs 0.5625 , -4.2%). Take $\sigma_{pr}(\text{sum}) \sim 0.05$ – 0.1 (tight; Brenner universal), $\sigma_{pr}(\text{split } \theta_2) \sim 0.15$ (wide; unknown), $\sigma_{pr}(b_1) \sim 0.05$, b_3 shrinkage ~ 0.05 .

7.2 Init chain (far-field, $\bar{h}_{init} \approx 22$)

$c(22; \theta_0) = 1 + \frac{9/8}{21} = 1.0536$ vs true $c_\perp(22) = 1.053$ ($+0.06\%$). Each station is therefore $\approx a_{nom}$ to $< 0.5\%$, all transient (absorbed in the hold):

$$\hat{a}_{x,init} = \frac{a_{nom}}{c(\bar{h}_{init}; \theta_0)} = 0.949 a_{nom}, \quad K_{h,init} = \left. \frac{1}{c} \frac{dc}{d\bar{h}} \right|_{\theta_0} = -2.4 \times 10^{-3} (\approx 0), \quad \sigma_{\delta h,init}^2 = 4k_B T \frac{a_{nom}}{c_\perp(\theta_{0z})}.$$

7.3 $P(4,4)$ init compression (CR-3 hazard)

The naive consistency map $P_{44}^{init} = (\partial a / \partial \theta|_{init})^2 \text{Var}(\theta_0)$ is dangerous *near wall* ($\sim 2000 \times$ DARE $\Rightarrow$ a-slot blow-up; buffer-deeper: $10a_{nom}^2 = 2.16 \times 10^{-3}$ diverges). But at far-field init $\partial a / \partial \theta|_{22} = \phi(22) \approx 0.045 a_{nom}$ is small, so

$$P_{44,naive}^{init}(22) = \phi(22)^2 \sigma_{pr}^2 \approx (0.045)^2 a_{nom}^2 (0.15)^2 \approx 1 \times 10^{-8} \ll P_{44}^{\text{DARE}} \sim 10^{-6}$$

so the naive map is **benign at far-field init** (unlike the near-wall case the audit flagged). Since steady-state P is init-independent (Task-04: P_{init} has no bias effect), this is a *stability*/transient concern only. Design rule:

Init table ($\kappa \in [1, 10]$, but $\kappa \lesssim 3$ safer — buffer-deeper marginal band 2 – $20 \times$ DARE; 2.16×10^{-7} stable, 2.16×10^{-3} blow-up):			
station	formula	far-field ($\bar{h}=22$)	residual
$\hat{a}_{x,init}$	$a_{nom}/c(\bar{h}_{init}; \theta_0)$	$0.949 a_{nom}$	$< 0.1\%$
$K_{h,init}$	$\frac{1}{c} dc/d\bar{h} _{\theta_0}$	-2.4×10^{-3}	≈ 0
$\sigma_{\delta h,init}^2$	$4k_B T a_{nom}/c_\perp(\theta_{0z})$	$4k_B T (0.949 a_{nom})$	$< 0.5\%$
P_{44}^{init}	$\max(\phi^2 \sigma_{pr}^2, \kappa P_{44}^{\text{DARE}})$	$\kappa P_{44}^{\text{DARE}} \sim 10^{-6}$	transient
$\theta_0, P_{\theta,0}$	$(\frac{9}{8}, 0)_z / (\frac{9}{16}, 0)_{xy}; \sigma_{pr}^2$	center; sum tight/split wide	4–5%

7.4 Warm-up sequence + FF-enable

Three phases on $\text{osc}_{1\text{Hz}}$: **(i) hold** $[0, 0.5\text{s}]$: existing G1 ($t < t_{\text{warmup}, kf} = 0.2\text{s}$); RLS trains at far field (a_nom-scale, G6 anchor); FF disabled ($\hat{a}'_x := 0$, A-floor). **(ii) descend**: FF-enables at the S6.4 criterion, with hysteresis. **(iii) osc** steady state.

FF-enable (with hysteresis). enable when $(\partial a' / \partial \theta)^2 P_\theta < (r_{\text{target}} \hat{a}'_x)^2$; disable only if $> (2r_{\text{target}} \hat{a}'_x)^2$. The factor 2 prevents chattering under the 46% chi-sq fluctuation (a single-threshold gate would toggle every few steps near the boundary). **Requirement:** enable *before* $\bar{h} < 3$ (else the no-FF ramp lag $\sim 30\%$, C1-12).

Expected enable time: $\hat{a}'_x \rightarrow 0$ far field makes the RHS tiny during hold, so enable fires on the descent once $\hat{a}'_x$ (hence RHS) has grown and P_θ collapsed; order $\sim 0.3\text{--}0.7\text{s}$ into the descent ($\bar{h} \sim 8\text{--}12$), comfortably before $\bar{h} < 3$. Flag: L1-measurable (RLS collapse rate $\times$ descend ϕ ramp).

7.5 A-floor BIBO + Q_{44} never zero

During $\hat{a}'_x = 0$ the slot-4 predict is $\text{AR}(1) \lambda_c a_x + (1 - \lambda_c) a_{x_{\text{det}}}^{\wedge}$ (fit still converging) or RW frozen a . Boundedness: $\text{AR}(1) e_{a_x}$ is bounded by $|e_{al}|_{\text{warmup}} (< 0.5\% \text{ far field, growing near wall} \Rightarrow \text{the enable-before-}\bar{h} < 3 \text{ requirement})$; RW by the no-FF ramp lag ($\sim 30\%$ near wall if descent finishes first). The **worst case** (C1-12: literal A-floor $Q_{44} = 0 + \text{G3 near wall} \rightarrow 170\text{--}490\%$) is ruled out:

Q_{44} never zeroed. The A-floor zeros $\hat{a}'_x$ for the *FF predict increment* only — **not** Q_{44} . Keep $Q_{44} \geq Q_{44}^{fl} > 0$ (a warm-up floor) so the a-slot stays adaptive; else A-floor $Q_{44} = 0 + \text{G3}$ (both channels dead) freezes the slot (170–490%, C1-12).

7.6 Numeric anchors

$\bar{h}_{\text{init}} = 22$ init residuals $< 0.5\%/\text{station}$ (see §7.3 box), all decayed within the 0.5s hold. FF-enable window $\sim 0.3\text{--}0.7\text{s}$ (descend, $\bar{h} \sim 8\text{--}12$). A-floor worst-case: $\text{AR}(1) \lesssim 0.5\%$ (far) / $\lesssim |e_{al}(3)|$ if late-enable; RW $\sim 30\%$ if enable misses $\bar{h} < 3$ — hence the hard requirement. $P_{44}^{\text{init}} = \kappa P_{44}^{\text{DARE}}$, $\kappa \lesssim 3$.

8. Independence — the physical g-loop

S3–S7 treated e_{al}/e_K as exogenous. Cross-session (2026-07-07) proved algorithmic acyclicity does *not* license this: the loop closes through physics. S8 analyzes the full loop, states the conditions for the exogenous treatment, and specifies the verifying tests.

8.1 The full loop map

PHYSICAL leg (uncuttable — it is the plant): $\hat{a}_x \xrightarrow{f_d} \delta x \rightarrow \hat{\sigma}_{\delta x_r}^2 \rightarrow a_{xm}$. **ALGORITHMIC** return (S6.5 severs algebraic shortcuts): a_{xm} feeds two paths back to $\hat{a}_x$,

$$\begin{cases} \text{P1: } y_2 \rightarrow \text{EKF slot 4} \rightarrow \hat{a}_x & (\text{fast, per-step}) \\ \text{P2: } \text{RLS} \rightarrow \hat{\theta} \rightarrow a(\bar{h}_d; \hat{\theta}), \text{FF}, Q_{44}/R_{22} \rightarrow \hat{a}_x & (\text{slow}) \end{cases}$$

The forbidden-inputs list cuts algebraic shortcuts (selfdet-type), but the physical leg cannot be cut by code discipline.

8.2 Mean-path (DC) analysis

Compose stage log-gains at $g=1$: $d \ln g / d \ln \hat{a}_x = -1$; $d \ln a_{xm} / d \ln g = dm/dg|_1 = -0.41$ (C1-08) $\Rightarrow$ forward physics $d \ln a_{xm} / d \ln \hat{a}_x = +0.41$. With the anchor $\rightarrow \hat{a}_x$ DC transfer (EWMA a_{m_det} : 1; RLS: $|P_{tilt}| \approx 0.9$) and lumped $\beta/\text{lag}/T$ attenuation $\eta \approx 0.76$:

$$S_{DC} = |dm/dg|_1 \cdot (\text{anchor DC}) \cdot \eta = \begin{cases} 0.41 \cdot 1 \cdot 0.76 \approx 0.31 & (\text{EWMA; matches parallel } S \approx 0.31) \\ 0.41 \cdot 0.9 \cdot 0.76 \approx 0.28 & (\text{RLS, smaller}) \end{cases}$$

$|S_{DC}| < 1 \Rightarrow$ unique contracting fixed point (stable). The C1-08 amplification $1/(1 - S_{DC})$ is the **mean equilibrium shift**, not instability: $1/(1-0.41)=1.69$ (the $|dm/dg|$ bound), measured $1/(1-0.31)=1.45$ (production 1.38–1.52). The +12% decomposes as open-loop 1.6–3.7% + ~ 8 pt loop-equilibrium shift; the linear $1/(1-S)$ captures leading order, the remainder is $m(g)$ nonlinearity.

8.3 Fluctuation / escape

DC contraction does *not* preclude noise-triggered escape. The chi-sq stream ν (rel. std 46%/step, $\tau_{int} \approx 39$) enters $\hat{\sigma}_{\delta x_r}^2 \rightarrow a_{xm} \rightarrow$ anchor every step. Through the *nonlinear* $m(g)$, a rare large downward a_{xm} excursion pushes $\hat{a}_x$, shifting g into a region where the local $|dm/dg|$ momentarily exceeds the contraction margin; the per-step anchor gain α sets how much of each excursion commits to $\hat{a}_x$. Below the cliff excursions decay; above, a large deviation escapes the finite basin ($S_{DC} < 1$ is only *local*). Seed-dependence reflects the rare trigger: $\alpha^*=0.008$ ($0.007 \rightarrow 0/5$, $0.008 \rightarrow 1/5$, $0.010 \rightarrow 3/5$, $0.020 \rightarrow 5/5$). This is metastable escape, not mean instability. The operative fluctuation-safety condition is S6 Ruling 4: $\alpha_{eq} \leq \alpha^*/M$, $M \geq 4$ (margin sized to clear the seed-dependent onset 0.007–0.008 under the 46% fluctuation; $M=1$ would sit at the 1/5-escape point).

8.4 The anchor-gain spectrum

One phenomenon, three loop gains (per-step anchor gain vs cliff $\alpha^*=0.008$):			
anchor	per-step gain	vs cliff	outcome
selfdet ($a_{x_{det}} := \hat{a}_x$)	1 (algebraic)	125× cliff	3/3 diverged
$a_{m_{det}}$ EWMA	0.005	1.6× below	marginal; +12% (~ 8 pt loop)
θ -RLS ($\lambda_f=0.999$)	9×10^{-4}	9× below	predicted safe
The three historical outcomes are the <i>same</i> loop at three gains — selfdet, $a_{m_{det}}$, and θ -RLS differ only in α .			

8.5 Shared-noise covariance

e_K (via s_K) and e_{a_x} (via K_{44}) are both driven by the *same* chi-sq stream ν (the a_{xm} noise, audit n_a): e_{a_x} gets $K_{44}\nu$ (P1), $s_K = (\partial a' / \partial \theta) K_\theta \nu$ (P2). Over the τ_{int} window,

$$\text{Cov}(e_K, e_{a_x}) \sim (\partial a' / \partial \theta) K_\theta K_{44} \text{Var}(\nu) \tau_{int}, \quad |\text{Cov}| \leq \rho_\nu \sqrt{\text{Var}(s_K) \text{Var}(e_{a_x})}.$$

Both anchor gains are small ($K_\theta \sim (1-\lambda_f) \sim 10^{-3}$, $K_{44} \sim 10^{-3}$ near wall), so ρ_ν is a product of two tiny gains: the cross is second-order. Consequence: treating e_K, e_{a_x} independent in the S3 carrier bookkeeping makes P *optimistic* by $2 \text{Cov}/Q_{44} \lesssim 5\%$ of Q_{44} (order estimate); identically for $\text{Cov}(s_{lvl}, e_{a_x}) \ll$ the 1% budget (C1-09).

8.6 The conditioned-exogeneity license

Ruling 5 (conditioned exogeneity). e_{al}/e_K may be treated as exogenous in the S3–S7 bookkeeping **iff** all four hold: (i) forbidden-inputs list (S6.5) — *algorithmic* layer; (ii) anchor-loop gain Ruling 4, $\lambda_f \geq 0.999$ — *fluctuation* layer; (iii) shared-noise Cov within the §8.5 bounds — *bookkeeping* layer; (iv) verified by the §8.7 ablations. Violation of *any* $\Rightarrow$ the S3 carrier table is invalid (the loop is not exogenous; the Q/R sizing is wrong).

8.7 Verification spec (interventions, L2)

- **(a) freeze- θ oracle:** same seed + `clear`; live RLS vs θ frozen at its converged value. The difference isolates the supplier-loop contribution. *Holds*: difference $\ll$ budget. *Fails*: difference grows/diverges.
- **(b) cut-physics:** $a_{ctrl} :=$ true gain (oracle) while the estimator runs normally — kills the physical leg, isolates the measurement chain. *Holds*: $\hat{a}_x$ error drops to open-loop 1.6–3.7% (confirms the ~ 8 pt was loop-driven).
- **(c) cut- y_2 :** y_2 permanently gated — isolates P2 (RLS) from P1. *Holds*: RLS-only path stays bounded; *Fails*: escape persists with y_2 off (RLS-loop unsafe).
- **(d) $\lambda_f \in \{0.99, 0.999\}$ cliff-transfer pair:** S6 Ruling 4 predicts 0.99 escapes (above cliff), 0.999 safe. Confirms/falsifies the $\alpha_{eq} \leftrightarrow \text{cliff}$ mapping.

8.8 p_{md} future channel

p_{md} (the LP mean $\delta \bar{x}_m$) carries $p_{md} \approx (1-g) D[k] + \text{noise}$, $D[k] = p_d[k+1] - \lambda_c p_d[k] - (1-\lambda_c) p_d[k-d]$ (known trajectory functional, $D=0$ at hold). This is a *signed first-moment* g -observation, currently discarded. It would de-confound g (scale) from θ (shape) — a *direct* g estimate independent of the a_{xm} scale, breaking the confound that Ruling 5 must otherwise bound. Out of MVP: needs its own noise/bias analysis (D -SNR, the $(1-g)$ linearization, control-loop interaction). Forward-ref T3.

9. Numeric anchors

Single lookup for the implementation. Fit floors: *L0 basis oracle* (`compute_L0.py`, 2026-07-06; $\text{osc}_{1\text{Hz}}$, $\bar{h} \in [1.2, 3.42]$, dwell-weighted LS).

9.1 L0 basis floors (frozen G0 bases)

Level = $a_{\text{model}}/a_{\text{true}} - 1$, slope = $a'_{\text{model}}/a'_{\text{true}} - 1$ (%). Dwell fit; gated column = fit trained on $\bar{h} \geq 1.5$, extrapolated to the trough.

metric	@1.2	@1.5	@3.42	RMS	gated@1.2
Z2 level (z)	−0.94	+0.12	−0.12	0.30	−1.83 [†]
Z2 slope (z)	+1.22	+0.92	−1.44	1.11	—
M2 level (xy)	+1.16	−0.78	+0.34	0.47	+5.40
M2 slope (xy)	−15.85	−1.31	+2.12	6.29	~21

Fallbacks: Z1 level −4.20 (dwell) / −2.67 (rel-LS) / −4.65 (gated) @1.2, slope 2.78 max; M1 level 5.16 max, slope 32.9 max. β (h-blur, z): Taylor +2.58@1.2 / +5.06@1.11; full-kernel ($\times 1.36$) +3.51 / +6.88. Far field ($\bar{h} > 5$): Z2/M1/M2 level < 1% (over-inflation gates clean); X0 retired (slope $\rightarrow +40\%$). **Binding floor everywhere** = **near-wall SLOPE** (1.4% z / 16% xy): no static basis reaches the 1% slope budget near wall. [†]Z2 gated −1.83@1.2 (−0.34@1.5) = `L0_summary.json results.Z2.W3_gated` ($\theta_1=1.078$, $\theta_2=0.091$); the `L0_tables.md` prose omits the Z2 gated row only.

9.2 Design constants

Every implementation number, with its defining section/ruling.

constant	value	source
forgetting λ_f	≥ 0.999 (working range $[0.999, 1)$)	Ruling 4 (§6.3)
cliff margin M	≥ 4	§6.3, §8.3
escape cliff α^*	0.008 (per-step anchor gain)	§6.3, §8.3
tilt projection $ P_{\text{tilt}} $	≈ 0.9 (−0.92 trough, +0.88 far)	§6.3
KF gain K_{44}	2.4×10^{-3} trough / $\sim 5 \times 10^{-3}$ flank / 5.8×10^{-9} far	§3.7, §5.7
sinusoid rejection A_{rej}	≈ 0.80 (flank-evaluated)	§3.4
window-centre lag L_{eff}	$19 = (1 - a_{\text{cov}})/a_{\text{cov}}$	§6.1
P_{44}^{init} scale κ	$\lesssim 3$ in $\max(\phi^2 \sigma_{pr}^2, \kappa P_{44}^{\text{DARE}})$	§7.3
train-band cap h_{train}	$\sim 4\text{--}5$	§6.2
FF-enable hysteresis	on at r_{target} , off at $2r_{\text{target}}$; before $\bar{h} < 3$	§7.4
θ_0 (prior centre)	$(\frac{9}{8}, 0)_z$ / $(\frac{9}{16}, 0)_{xy}$	§7.1
prior widths σ_{pr}	sum 0.05–0.1, split 0.15, b_1 0.05	§7.1
R_{RLS}	$K_{\text{var}} IF_{\text{eff}} (\hat{a}_x + \xi)^2$	§6.2
gate-off P_{44} floor	$[b_{\text{vl}}^{\text{ex}}]^2$ (AR1) / $[e_K dh']^2$ (RW)	Ruling 3 (§5.4)
warm-up Q_{44} floor	> 0 (never zeroed)	§7.5
Q_{44} ramp inflation	rejected ($e_K \Delta h_d$ is bias)	Ruling 1 (§3.3)
relative-form rule	Δh_d^2 terms $\propto \hat{a}_x'^2(\bar{h})$	Ruling 2 (§3.6)
exogeneity license	4 conditions (i)–(iv)	Ruling 5 (§8.6)

9.3 Flag inventory (L1–L3 must resolve)

Every inline hypothesis/estimate in S3–S8, with its resolving test — the bridge to the verification matrix.

#	flag / hypothesis	§	resolving test
i	6.4×10^{-5} $O(4)$ prefactor is amplitude-dependent	3.3	L2 amplitude sweep
ii	flank $K_{44} \sim 5 \times 10^{-3}$ inferred; $A_{rej}=0.80$ flank-evaluated	3.4	fig_L2_eax_channels
iii	RW trough sensitivity $0.46r$ vs audit $\sim 1.0r$ ($2 \times$ gap)	5.3	L2 gain overlay
iv	$\alpha_{eq} \leftrightarrow$ cliff transfer is a HYPOTHESIS	6.3	$\lambda_f \{0.99, 0.999\}$ pair (L3)
v	$ P_{tilt} \approx 0.9$ borrowed (G6/para-c); exact Z2/M2 pending	6.3	L0/L2
vi	$\eta \approx 0.76$ back-solved lumped factor	8.2	L2 decomposition
vii	$\text{Cov} \leq 5\%$ Q_{44} order estimate	8.5	L2
viii	FF-enable 0.3–0.7 s order estimate	7.4	L1